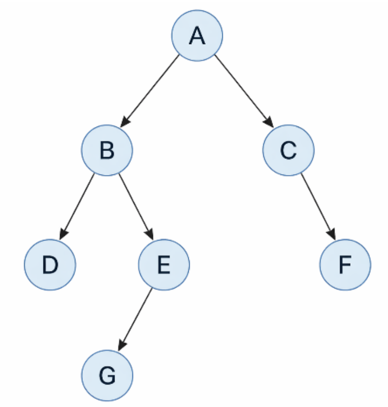
Module Description: Artificial Intelligence

Module Code: CMPG 313

Practical Assessment: Uninformed search using python

Consider the following graph



Instructions:

Consider .py scripts for BFS, DFS, and ID-IDF attached to this lab. Furthermore, follow these instructions to complete the practical lab.

1. Open the file **uninformed_test.py** and edit the graph definition so that it correctly represents the tree shown in the provided diagram as an adjacency list (dictionary). Ensure that:
    - all nodes in the diagram are included,
    - nodes with no children are represented correctly, and
    - the parent-child relationships in the diagram are preserved accurately.
    - Do not modify any other part of the program.
2. Update the function calls so that each search algorithm is provided with the correct inputs. Ensure that:
    - all algorithms use the constructed graph as input,
    - the traversal begins from the correct starting node (the root of the tree), and
    - for Iterative Deepening Depth-First Search (IDDFS), an appropriate **maximum depth limit** is specified based on the depth of the tree.
3. Submit a code output for **uninformed_test.py** and further attempt **Practical Assignment [MCQ] - Lab 4**